

Brief Report on US20170371329A1: Multi-modal Sensor Data Fusion for Perception Systems

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Abstract

This report summarizes patent US20170371329A1, explains its core technical problem, architecture, relevance to a distributed field intelligence system, and its limitations. The report is intended for a short technical presentation or project literature survey.

1 Basic Identification

- **Publication number:** US20170371329A1
- **Title:** Multi-modal sensor data fusion for perception systems
- **Original assignee:** United Technologies Corporation
- **Current assignee listed by Google Patents:** RTX Corporation
- **Publication date:** December 28, 2017
- **Status noted by Google Patents:** granted through related version US11422546B2

2 Core Problem

This patent addresses a central issue in perception systems: different sensors observe the same environment through different modalities and often in different coordinate systems. A video camera provides image data, LiDAR provides range/depth information, radar may provide distance and velocity, and GPS gives global position. To make reliable perception decisions, the system must fuse these streams into a coherent representation.

A key difficulty is *registration*: sensor data from different modalities must be aligned into a common coordinate system or representation. Classical approaches may require manually designed anchor points or separate feature extraction pipelines for each modality. This patent proposes a deep-learning-based approach to dynamically fuse multimodal sensor streams and detect useful perception indicators.

3 Main Technical Pipeline

The method can be summarized as:

Multi-modal sensor data → fusion → region of interest detection → patch extraction → deep convolutional neural network → post-processing → perception indication

More concretely, the system:

1. receives multimodal sensor data from a plurality of sensors;

2. detects at least one region of interest in the fused data;
3. detects patches of interest based on the detected regions;
4. applies a deep convolutional neural network to the patches;
5. post-processes the model output;
6. outputs a perception indication.

The patent gives examples such as edge detection, occlusion-edge detection, auto-registration, sensor misalignment detection, and alignment adjustment.

4 Relevance to Field Intelligence

This patent is highly relevant to the **perception layer** of a field intelligence system. A field device may have a camera, microphone, LiDAR, thermal sensor, IMU, GPS, or radar. Raw data from these sensors is not enough; the device must fuse them into a single world-state estimate.

For example, a field device should not maintain separate unconnected facts such as:

“RGB camera sees a shape; thermal sensor sees heat; LiDAR sees a cluster.”

Instead, it should fuse them into an entity-level interpretation:

“Entity E17 is likely a person, located 8.2 meters ahead, partially occluded, confidence 0.89.”

This patent is therefore useful as prior art for multimodal perception, but it does not by itself solve mission reasoning, multi-device coordination, or asynchronous field-data synchronization.

5 Limitations

The patent is mainly about perception fusion. It does not deeply address:

- mission-conditioned reasoning;
- distributed field-device coordination;
- operator guidance;
- asynchronous logging;
- SLAM or semantic mapping as a complete system.

6 Conclusion

US20170371329A1 should be read as a perception-layer patent. Its main value for a field intelligence project is its treatment of multimodal sensor fusion and conversion of heterogeneous sensor streams into perception indications. It is important prior art for the device’s ability to understand what is present in its environment, but it must be combined with separate mechanisms for mission reasoning, spatial mapping, and distributed situational awareness.

Reference

Google Patents, “US20170371329A1 - Multi-modal Sensor Data Fusion for Perception Systems.”
<https://patents.google.com/patent/US20170371329A1/en>